MIL-S-[illegible] (AR)
7 July 1977
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

SWITCHING UNIT, SAFING, XM70

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of switching unit entitled Switching Unit, Safing, XM70 which is used with the Sprint warhead.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-6872	- Soldering Process, General Specification for.

STANDARDS

FEDERAL

FED-STD-209	- Clean Room and Work Station Requirements, Controlled Environments.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1135

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DRAWINGS

ORDNANCE CORPS

9276931	- Spring, Helical Torsion.
9276947	- Roller Arm Assembly.
9276949	- Intermediate Gear Assembly.
9276950	- Index Gear Assembly.
9276953	- Actuator Retainer Assembly.
9276954	- Actuator and Shaft Assembly.
9276956	- End Plate and Hub Assembly.
9276959	- Motor, Torque, DC.
9276963	- Housing Assembly.
9276965	- Contact Plate Assembly.
9276966	- End Plate Assembly.
9276970	- Housing.
9276972	- End Plate Assembly.
9276975	- Contact Plate Assembly.
9276976	- End Plate and Hub Assembly.
9276983	- Motor and Housing Assembly.
9276984	- Housing and Actuator Assembly.
9277000	- Switching Unit, Safing, XM70.
9278227	- Packing and Marking for Bag, Packing for Switching Unit, Safing, XM70.
EL-9276931	- Equipment List.
EL-9276949	- Equipment List.
EL-9276950	- Equipment List.
EL-9276953	- Equipment List.
EL-9276954	- Equipment List.
EL-9276956	- Equipment List.
EL-9276959	- Equipment List.
EL-9276963	- Equipment List.
EL-9276965	- Equipment List.
EL-9276970	- Equipment List.
EL-9276975	- Equipment List.
EL-9276976	- Equipment List.
EL-9276983	- Equipment List.
EL-9276984	- Equipment List.
EL-9277000	- Equipment List.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1	- Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The switching unit shall comply with all requirements specified on the drawing (dwg) 9277000, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein. Assembly shall be performed in a Class 10,000 clean room, in accordance with FED-STD-209.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-R, Dover, New Jersey 07801. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Insulation resistance.

3.6.1 Reset (safe) position. The insulation resistance between terminals A, B, C, D, I, J, K, M, P, S and case shall be 100 megohms minimum. Insulation resistance measurements shall be made between each isolated terminal shown above and all other terminals and case connected to a common electrical point. Terminals I or J shall not be measured or tested relative to each other due to the existence of an internal resistor, see Figure 1. Measurements shall be made upon application of 500 plus or minus (±) 50 volts direct current (vdc) for 5 seconds minimum when tested as specified in 4.4.1.

3.6.2 Set (arm) position. The insulation resistance between terminals A, C, I, K, L, M, P, S and case shall be 100 megohms minimum. Insulation resistance measurements shall be made between each isolated terminal shown above and all other terminals and case connected to a common electrical point. Measurements shall be made upon application of 500 ± 50 volts direct current (vdc) for 5 seconds minimum when tested as specified in 4.4.1.

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3.7 Electrical contacts.

3.7.1 Circuit resistance. The circuit resistance of each closed circuit shall not exceed the limits shown in Table I when tested as specified in 4.4.2.1.

3.7.2 Current carrying capability. Switches shall be capable of carrying a 12.5 ± 5 ampere direct current resistive load for 60 ± 1 seconds minimum at 35 ± 1 vdc open circuit when tested as specified in 4.4.2.2. Table I lists the terminals involved. Afterwards, the unit shall comply with 3.7.1 (Set position only).

3.7.3 Contact chatter. Where testing for contact chatter is required, no opening of a closed switch or closing of an open switch shall exceed 5 microseconds, single event, when tested as specified in 4.4.2.3. For this test, contact load shall be 100 ± 5 milliamps dc (limited to 6 ± 0.5 vdc open circuit). Unless otherwise specified, all contact pairs in Table I shall be monitored, regardless of the unit's position. Contact chatter is defined as a change in contact resistance through 50 ± 5 ohms. See Figure 1 for circuit diagram.

Table I

Set (Arm) Position		Before Environmental Testing	After Environmental Testing	Test Code No. (See 3.5 and Table II)
Lead 1	Lead 2	Reading	Reading	
A	B	30 milliohms	50 milliohms	6101--
C	D	30 milliohms	50 milliohms	6102--
I	J	30 milliohms	50 milliohms	6103--
P	N	80 milliohms	100 milliohms	6104--
Reset (Safe) Position				
K	L	30 milliohms	50 milliohms	6105--
M	N	80 milliohms	100 milliohms	6106--
I	J	36.8 - 38.0 Kilo-ohms	36.8 - 38.0 Kilo-ohms	6107--
S	T	11.4 - 14.6 ohm	11.4 - 14.6 ohm	6108--

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Table II

Test Condition	Last Two Digits
Before Environments (see 3.7.1)	00
Flight Vibration (see 3.11.1.2)	01
Random Vibration (see 3.11.2)	02
Operational Shock (see 3.10.2)	03

3.8 Seal integrity. The unit shall be subjected to a helium leak test. The total leak rate shall not exceed $1x10^{-5}$ atm cc He/sec when tested as specified in 4.4.3.

3.9 Operational.

3.9.1 Switch transfer. With the unit at ambient temperature (80 ± 10 degrees Fahrenheit (°F)), automatically apply a step input of 18.0 +0.0, -0.5 vdc to "set" the unit (- to pin S, + to pin M) using circuits K-L and I-J to indicate switch transfer. (See Figure 1). The time from application of 18 vdc to the closing of circuit I-J shall be not less than 30 milliseconds (msec) nor more than 81 milliseconds. Return the unit to the reset position (+ to pin S, - to pin P) with 25 ± 7 vdc, then apply a step input of 32.0 +0.5, -0.0 vdc to set the unit. (See Figure 1). Current shall not exceed 3.0 amperes nor be applied for more than 100 milliseconds of operation. Switch transfer time observed shall be within the 30 to 81 milliseconds time frame. Testing shall be in accordance with 4.4.4.1. Use test code numbers 8101-- (for step input of 18 vdc). See 3.5 and Table II.

3.9.1.1 Switch transfer (life cycle). With the unit at ambient temperature (80 ± 10°F), automatically apply a step input of 18.0 +0.0, -0.5 vdc to "set" the unit (- to pin S, + to pin M) using circuits K-L and I-J to indicate switch transfer (see Figure 1). The time from application of 18 vdc to the closing circuit I-J shall be not less than 30 milliseconds nor more than 90 milliseconds. Return the unit to the reset position (+ to pin S, - to pin P) with 25 ± 7 vdc. Current shall not exceed 3.0 amperes and shall be applied for not more than 100 milliseconds of operation. Testing shall be in accordance with 4.4.4.1. Use test code number 8102--. See 3.5 and Table II.

3.9.2 Voltage discrimination. With the unit at ambient temperature (80 ± 10°F) and in the reset (safe) condition, automatically apply a step input (- to pin S, + to pin M) of 10.0 + 0.0, - 0.5 vdc for 100 milliseconds as per 4.4.4.2. The device shall not change state nor transfer any switches.

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3.9.3 Voltage sensitivity. With the unit at ambient temperature (80 ± 10°F) and in the safe condition, automatically apply a step input (- to pin S, + to pin M) of 17.5 vdc for 100 milliseconds, per 4.4.4.3. If the device transfers in 80 msec or less, reset the device using 25 ± 7 vdc and apply 1.0 volt less. Repeat this process until the device fails to transfer in less than 80 msec. Record the lowest voltage at which an 80 msec transfer was achieved. Use test code 8301-- (for lowest 80 msec transfer voltage). See 3.5 and Table II.

3.9.4 Repeatability. With the unit at ambient temperature (80 ± 10°F), automatically apply a step input of 18.0 +0.0, -0.5 vdc to "set" the unit (- to pin S, + to pin M) using circuits K-L and I-J to indicate switch transfer. (See Figure 1). The time from application of 18 vdc to the closing of circuit I-J shall be not less than 30 milliseconds nor more than 81 milliseconds. Return the unit to the reset position (+ to pin S, - to pin P) with 25 ± 7 vdc. This set (arm) and reset (safe) constitutes one operation. The switch transfer shall be performed 5 times for a total of 5 operations. The five operating times (set only) for a particular unit when taken as a group, shall be repeatable within 5 milliseconds (absolute), or less when tested as specified in 4.4.4.4. Use test code numbers 8401--, 8402--, 8403--, 8404-- and 8405--. See 3.5 and Table II.

3.10 Mechanical shock.

3.10.1 Transportation shock. The unit in the reset (safe) position shall be subjected to a total of 6 shocks (one in each direction along each of three mutually perpendicular axes) and shall not change state or transfer any switches when tested as specified in 4.4.5.1. A half sine wave form of 30 gravity units (g's) ± 3 g's and 11 ± 1.0 milliseconds duration shall be used. After completion of this test, there shall be no evidence of damage (see 6.5), and the unit shall comply with the requirements of 3.6.1, 3.7.1 (reset position only), 3.8 and 3.9.1.

3.10.2 Operational shock. The unit in the set (arm) position shall be subjected to a total of 12 shocks (two different shocks in each direction along each of three mutually perpendicular axes) and shall not change state or transfer any switches. The shocks in each direction using the values from Table III shall be as defined in MIL-STD-810, Method 516.1, Figures 516.1-1 and 2. Testing shall be performed per 4.4.5.2, and chatter shall be monitored per 3.7.3.

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Table III

Shock Sequence	MIL-STD-810 Figure	Duration	Amplitude
1st Shock	516.1-1 (Term. Peak)	6 ms	100 g
2nd Shock	516.1-2 (Half Sine)	3 ms	300 g

After completion of this test, there shall be no evidence of damage (see 6.5) and the units shall comply with 3.6.2 and 3.7.1. (Set position only).

3.11 Vibration.

3.11.1 Sinusoidal vibration.

3.11.1.1 Transportation. The unit in the reset (safe) position, shall not change state or transfer any switches when vibrated for resonance search at the frequencies shown in Table IV, sweeping logarithmic from 5 to 500 to 5 Hz in 15 ± 1 minutes, in each of three mutually perpendicular axes. Test per 4.4.7. After completion of this test, test for compliance with 3.6.1 and 3.7.1 (reset position only). There shall be no evidence of damage (see 6.5).

Table IV

Frequency Hertz (Hz)	Magnitude
5-27	1.3 g
27-52	0.036 in. da
52-500	5 g

3.11.1.2 Flight. The unit in the set (arm) position shall be subjected to a resonance search by being vibrated in each of three mutually perpendicular axes at 5 to 20 Hz at 0.1" da, and 20 to 200 Hz at 2 g's maximum, sweeping the significant resonances. Dwell no more than 15 seconds at each resonance discovered to confirm the existence of the resonance, (but dwell at no more than the four most significant resonances in each plane). Following this, make a 30 second logarithmic sweep of 5 to 280 Hz in each of three mutually perpendicular axes at the magnitudes listed in Table V. Monitor functional contacts for chatter per 3.7.3. After completion of this test, test for compliance with 3.6.2 and 3.7.1 (set position only). There shall be no evidence of damage (see 6.5). Testing shall be per 4.4.7.

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Table V

Frequency Hertz (Hz)	Magnitude
5-11	0.5 in. da
11-44	3.0 g (0 Peak)
44-100	4.0 g (0 Peak)
100-200	13.0 g (0 Peak)
200-280	5.0 g

3.11.2 Random vibration. The unit in the set (arm) position shall be subjected to the vibrations specified in Table VI for one minute in each of three mutually perpendicular axes without evidence of damage (see 6.5) and shall not change state or transfer any switches. During this test, monitor functional contracts per 3.7.3. Testing shall be as specified in 4.4.6. After completion, test for compliance with 3.6.2 and 3.7.1. (Set position only).

Table VI

Frequency Hertz (Hz)	Power Spectral Density	Tolerance
10-50	0.01 g^2/Hz	± 3 dB
50-550	0.1125 g^2/Hz	± 3 dB
550-1500	3.8 dB/Oct	± 3 dB
1500-2500	0.40 g^2/Hz	± 3 dB
2500-3000	0.45 g^2/Hz	± 3 dB

3.12 Acceleration. The unit in the set (arm) position shall be subjected to an acceleration of 170 g minimum for a duration of one minute minimum in each direction of the flight axis (xx) and 80 g minimum for one minute in each direction of axis YY and ZZ (see Figure 2) without changing state of transferring any switches. Testing shall be in accordance with 4.4.8. After completion of this test, there shall be no evidence of damage (see 6.5) and the unit shall comply with the requirements of 3.6.2 and 3.7.1 (set position only).

3.13 Low temperature. The unit, in the reset (safe) position, shall withstand exposure to a temperature of -40 ± 5°F for 4 hours minimum without damage (see 6.5) when tested as specified in 4.4.9. After completion of this test, the unit shall meet the requirements of 3.6.1, 3.7.1 (reset position only) and 3.9.1.

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3.14 High temperature. The unit, in the reset position, shall withstand 125 + 5°F for 4 hours minimum without damage when tested as specified in 4.4.10. After completion of this test the unit shall meet the requirements of 3.6.1, 3.7.1, (reset position only) and 3.9.1.

3.15 Temperature shock. The unit in the reset (safe) position, shall withstand three (3) temperature shock cycles. One cycle is defined as the exposure at 125 ± 5°F to -40 ± 5°F and back to 125 + 5°F without evidence of damage when tested as specified in 4.4.11. Upon completion of the test, the unit shall comply with the requirements of 3.6.1, 3.7.1 (reset position only), 3.8 and 3.9.1.

3.16 Life cycle. The unit shall be capable of at least 35 operations and function within the requirements of 3.9.1 and 3.9.2 when tested as specified in 4.4.12. After every 10th set cycle the unit shall be tested for compliance with 3.7.1, reset, and tested per 3.9.2. Repeat for 35th cycle.

3.17 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460 and MIL-S-6872.

3.18 Back emf. The torque motor back emf constant K_B, shall read 0.020 volts/radian/second minimum when tested in accordance with 4.4.13.

3.19 Workmanship.

3.19.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.19.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.19.3 Burrs. Parts shall be free from burrs.

3.19.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.19.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.19.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 5 complete assemblies (see dwg 9277000) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table VII. Tests shall be conducted in order specified in Table VII.

Table VII

First article inspection and tests

Inspection or Test	Test Unit No.'s All	1	2	3	4	5	Dest	Non-Dest
1. Classification of defects/characteristics (see 4.3.2.1)	X							X
2. Seal integrity (see 3.8)	X							X
3. Circuit resistance (see 3.7.1)	X							X
4. Insulation resistance (see 3.6)	X							X
5. High temperature (see 3.14)			X	X	X	X		X
6. Low temperature (see 3.13)			X	X	X	X		X
7. Transportation vibration (see 3.11.1.1)		X						X
8. Temperature shock (see 3.15)	X							X
9. Transportation shock (see 3.10.1)	X							X
10. Operational (see 3.9)	X							X
11. Life cycle (see 3.16)			X					X
12. Seal integrity (see 3.8)	X							X
13. Vibration, flight (see 3.11.1.2)	X							X

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Table VII (Cont'd)

Inspection or Test	Test Unit No.'s All	1	2	3	4	5	Dest	Non-Dest
14. Random vibration (see 3.11.2)	X						X	
15. Acceleration (see 3.12)	X							X
16. Operational shock (see 3.10.2)	X						X	
17. Current carrying capability (see 3.8.2)	X						X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor defects/characteristics table (see Table VIII and Table IX), and inspected for all defects/characteristics in the applicable classification. Any defects/characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of defects/characteristics in each classification need not be followed as long as all defects/characteristics inspections are performed.

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Table VIII

MIL-S-48896 (AR) Major Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IX
Minor Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of defects/characteristics (see 3.2). Critical, major and minor defects/characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Spring, helical torsion (see dwg 9276931 covering a detail of dwg 9277000).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Spring torques incorrect	Gage
	102. Direction of helix incorrect	Visual
	103. Overall length incorrect	Gage
Minor:		
	201. Evidence of poor workmanship (see 3.19)	Visual

4.3.2.1.2 Roller arm assembly (see dwg 9276947 covering a detail of dwg 9277000).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Roller does not move freely after assembly	Manual
	102. Any part missing or incorrectly assembled	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.19)	Visual

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4.3.2.1.3 Intermediate gear assembly (see dwg 9276049 covering a detail of dwg 9276984).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Staking missing or incorrect	Visual
102.	Staking does not comply with drawing note 2	Gage
103.	Rotating and breakaway torque incorrect	Gage
104.	Any part missing or incorrectly assembled	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.19)	Visual

4.3.2.1.4 Index gear assembly (see dwg 9276950 covering a detail of dwg 9276984).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Staking missing or incorrect	Visual
102.	Staking does not comply with drawing note 2	Gage
103.	Any part missing or incorrectly assembled	Visual
104.	Protrusion of flared material greater than specified on the drawing	Gage
Minor:		
201.	Evidence of poor workmanship (see 3.19)	Visual

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4.3.2.1.5 Actuator retainer assembly (after staking) (see dwg 9276953 covering a detail of dwg 9276984).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Flanged bearings loose in retainer (2 places) (see drawing note 2)	Gage
	102. Bearing inside diameter incorrect	Gage
	103. Any part missing	Visual
Minor:		
	201. Any raised material above surface (see drawing note 3)	Gage

4.3.2.1.6 Actuator and shaft assembly (see dwg 9276954 covering a detail of dwg 9276984).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Pin not flush to or below surface (see 6.6)	Visual
	102. Installed pin does not comply with drawing note 2	Gage
	103. Distance between shaft groove and actuator incorrect	Gage
	104. Angularity of actuator and shaft flats incorrect	Gage
Minor:	None defined.	

4.3.2.1.7 End plate and hub assembly (see dwg 9276956 covering a detail of dwg 9277000).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Terminal height incorrect (9 places)	Gage
	102. Staking missing or incorrect	Visual
	103. Staking does not comply with drawing note 3	Gage
	104. Hub torque incorrect after staking	Gage

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Major: Cont.

105. Soldering incorrect (see 3.17).....Visual

Minor:
201. Evidence of poor workmanship (see 3.19).....Visual

4.3.2.1.8 Housing assembly (after staking) (see dwg 9276963 covering a detail of dwg 9276983).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Flanged bearings loose in housing (2 places) (see drawing note 2)	Gage
102.	Bearing inside diameter incorrect (2 places)	Gage
103.	Threaded inserts not flush or below surface (4 places)	Gage
104.	Height of pin incorrect, (2 places)	Gage
105.	Dowel pin not below surface	Gage
106.	Height of groove pin incorrect (1 place)	Gage
Minor:		
201.	Any raised material above surface (see drawing note 3)	Gage
202.	Any pins loose, missing or bent	Visual/Manual

4.3.2.1.9 Contact plate assembly (see dwg 9276965 covering a detail of dwg 9277000).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any circumferential cracks or missing material in flared surfaces of eyelets (see 6.6)	Visual
102.	Eyelets do not conform to drawing note 2	Gage
103.	Orientation of offset contact incorrect	Visual
Minor:	None defined.	

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4.3.2.1.10 End plate assembly (prior to welding) (see dwg 9276966 or 9276972 as applicable covering a detail of dwg 9277000).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Any contact assembly loose after flaring	Manual
	102. Any circumferential cracks or missing material in flared surfaces (see 6.6)	Visual
Minor:	None defined.	

4.3.2.1.11 End plate assembly (after welding) (see dwg 9276966 or dwg 9276972 as applicable covering a detail of dwg 9277000).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Any flared terminal not welded	Visual
	102. Switch rotor assembly improperly installed	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.19)	Visual

4.3.2.1.12 Housing (see dwg 9276970 covering a detail of dwg 9276963).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Location of mounting holes incorrect (4 holes)	Gage
	102. Pitch diameter of threads incorrect (4 holes)	Gage
	103. Minor diameter of threads incorrect (4 holes)	Gage
	104. Depth of full thread incorrect (4 holes)	Gage

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Major: Cont.

105. Protective finish incorrect, scratched or marred exposing base material.....Visual

Minor:

201. Evidence of poor workmanship (see 3.19).....Visual

4.3.2.1.13 Contact plate assembly (see dwg 9276975 covering a detail of dwg 9277000).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any circumferential cracks or missing material in flared surfaces of eyelets (9 places) (see 6.6)	Visual
102.	Eyelets do not conform to drawing note 2	Gage
103.	Resistor missing or incorrectly installed	Visual
104.	Soldering incorrect (see 3.17)	Visual
105.	Orientation and quantity of contact assemblies and contact support incorrect	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.19)	Visual

4.3.2.1.14 End plate and hub assembly (see dwg 9276976 covering a detail of dwg 9277000).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Terminal height incorrect (4 places)	Gage
102.	Staking missing or incorrect	Visual
103.	Staking does not comply with drawing note 3	Gage
104.	Hub torque incorrect after staking	Gage
105.	Soldering incorrect (see 3.17)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.19)	Visual

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4.3.2.1.15 Motor and housing assembly (Prior to installing motor retainer) (see dwg 9276983).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Any screw missing or loose (4 places)	Visual/Manual
	102. Serial numbers of rotor stator and brush ring do not match	Visual
Minor:	None defined.	

4.3.2.1.16 Motor and housing assembly (see dwg 9276983).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Rotor pinion end play incorrect	Gage
Minor:		
	201. Evidence of poor workmanship (see 3.19)	Visual

4.3.2.1.17 Housing and actuator assembly (see dwg 9276984).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Actuator shaft end play incorrect	Gage
	102. Leads missing or lead color coding incorrect	Visual
	103. Any screw missing or loose (3 places both ends)	Visual/Manual
Minor:	None defined.	

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4.3.2.1.18 Switching unit, safing, XM70 (Prior to installing end plates) (see dwg 9277000).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Insulators missing (2 places)	Visual
	102. Wires (2) incorrectly installed	Visual
	103. Serialization improper or missing (see 3.4)	Visual
	104. Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.19)	Visual

4.3.2.1.19 D.C. torque motor (see dwg 9276959).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Torque motor back EMF incorrect (see 3.18)	Gage
Minor:	None defined.	

4.3.2.1.20 Switching unit, safing, XM70 (see dwg 9277000).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Overall length, width and height incorrect	Gage
	102. Location of terminal pins incorrect (13 places)	Gage
	103. Diameter of terminal pins incorrect	Gage
	104. Identification plate or labels missing, improperly located or installed	Visual
	105. Solder missing at screw heads (4 places)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.19)	Visual

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4.3.2.1.21 Preparation for delivery (see Section 5).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Packaging, packing and marking not in accordance with drawing	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.19)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Seal integrity (see 3.8)	Major
b.	Circuit resistance (see 3.7.1)	Major
c.	Insulation resistance (see 3.6)	Major
d.	Operational (see 3.9.1, 3.9.2 and 3.9.4)	Major
e.	Temperature shock (see 3.15)	Major
f.	Transporation vibration (see 3.11.1.1)	Major
g.	Transporation shock (see 3.10.1)	Major
h.	Random vibration (see 3.11.2)	Major
i.	Operational shock (see 3.10.2)	Major
j.	Current carrying capability (see 3.7.2)	Major

Of these tests, h, i and j are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Seal integrity, circuit resistance, insulation resistance, and operational. Each switching unit shall be tested in accordance with 4.4.3, 4.4.2.1, 4.4.1, 4.4.4.1, 4.4.4.2, and 4.4.4.4. If any item fails to comply with 3.8, 3.7.1, 3.6, 3.9.1, 3.9.2, and 3.9.4, the item shall be rejected and removed form the lot.

4.3.3.2 Temperature shock, transportation vibration, transportation shock, random vibration, operational shock, and current carrying capability. A sample shall be selected at random in accordance with Table X. The sample shall be tested in accordance with 4.4.11, 4.4.7, 4.4.5.1, 4.4.6, 4.4.5.2 and 4.4.2.2. If any item fails to comply with 3.15, 3.11.1.1, 3.10.1, 3.11.2, 3.10.2, and 3.7.2, the lot shall be rejected.

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Table X

Production Lot Acceptance

No. of Acceptable Units Required to be Delivered	No. of Sample Units to be Destructively Tested (Sample Size)	Inspection Total
1-8	8	9-16
9	9	18
10-18	10	20-28
19-28	11	30-39
29-57	12	41-69
58-429	13	71-442
430 and up	14	444 and up

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major defects/characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL's 2.1). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor defects/characteristics and tests. Acceptance inspection equipment for minor defects/characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Insulation resistance. The insulation resistance shall meet the applicable requirements of 3.6 (set, reset or both) when tested in accordance with MIL-STD-202, Method 302, Test Condition "B". The current shall be limited to 5 milliamperes.

4.4.2 Electrical contacts.

4.4.2.1 Circuit resistance. The test current shall be 100 ± 5 milliamperes with the open circuit voltage limited to 6.0 ± 0.5 vdc. The measured circuit resistance shall meet the requirements of 3.7.1 when measured at the pins in accordance with the four wire method of MIL-STD-202, Method 307.

4.4.2.2 Current carrying capability. With the unit in the set (arm) position, and using an approved constant current or current-limited source (resistive load) apply 12.5 ± 0.5 amperes dc to the terminals as listed in Table I for 60 ± 1 seconds. The open circuit voltage shall be 35 ± 1 volts dc. After this test, test for compliance with 4.4.2.1. (Set position only).

4.4.2.3 Contact chatter. Apply 100 ± milliamperes direct current (limited to 6 ± 5 vdc open circuit voltage) to the terminals of each circuit to be monitored. With equipment capable of detecting chatter on each circuit, measure the accumulative chatter during any 500 ± 50 microsecond interval. If the limits of 3.7.3 are violated, a failure indication shall be given and maintained until reset for that circuit.
NOTE: Do not monitor between terminals S and T.

4.4.3 Seal integrity. Using a mass spectrometer helium leak detector, the unit shall be tested at an absolute pressure of one torr or less for leakage in excess of 1 X 10^{-5} atm cc/sec of helium.

4.4.4 Operational.

4.4.4.1 Switch transfer. Automatically transfer the switch in accordance with the requirements of paragraph 3.9.1. Measure and record the transfer time and motor current.

4.4.4.2 Voltage discrimination. The unit shall not change state nor transfer when a step input of 10.0 + 0.0, -0.5 volts (dc or 60 Hertz AC - zero to peak) is applied. The unit shall comply with the requirements of paragraph 3.9.2.

4.4.4.3 Voltage sensitivity. Subjecting the switch to the voltage as specified in 3.9.3, determine and record the lowest voltage at which an 80 msec transfer time is achieved.

4.4.4.4 Repeatability. Automatically transfer the switch in accordance with the requirements of paragraph 3.9.4. Measure and record the set times only.

4.4.5 Shock.

4.4.5.1 Transportation shock. The unit situated in the reset or safe position shall be rigidly mounted onto a fixture cable of orientating the unit along each of three perpendicular axes relative to the shock source. The unit and fixture shall be secured to an approved shock machine and subjected to requirements of 3.10.1. After this test, the unit shall be examined in accordance with 4.4.1, 4.4.2.1, 4.4.3, 4.4.4.1 and inspected for damage (see 6.5).

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4.4.5.2 Operational shock. The unit shall be tested in accordance with the procedure of 4.4.5.1, except that: the unit shall be in the set or arm position when subjected to the requirements of 3.10.2, all circuits shall be monitored for chatter per 4.4.2.3, and after this test, the unit is to be examined for compliance with 4.4.1, 4.4.2.1 (set position only) and inspected for damage (see 6.5).

4.4.6 Random vibration. The unit, in the set or arm position shall be rigidly mounted to a fixture which is capable of being supported readily in each of three mutually perpendicular axes. The unit shall be vibrated over the frequency ranges of random conditions as defined in 3.11.2. All function circuits shall be continually moitored during the test per 4.4.2.3. After this test the unit shall be tested in accordance with 4.4.1 and 4.4.2.1 (set position only) and examined for damage (see 6.5).

4.4.7 Sinusoidal vibration. The unit in the specified position shall be rigidly mounted to a fixture which is capable of being positioned readily in each of three mutually perpendicular axes. The unit and fixture shall be secured to an approved vibration machine and shall be vibrated over the applicable fequency ranges of sinusoidal conditions specified in 3.11.1.1 or 3.11.1.2. Chatter monitoring as required by 3.11.1.2 shall be per 4.4.2.3. After each test, the unit shall be tested in accordance with 4.4.1, 4.4.2.1 and examined for damage (see 6.5).

4.4.8 Acceleration. Using a centrifuge, subject the unit to the linear acceleration specified in 3.12. The acceleration along each of the three mutually perpendicular axes must occur in both directions (6 accelerations total). After this test the unit shall be tested in accordance with 4.4.1, 4.4.2.1 and 4.4.4.

4.4.9 Low temperature. The unit in the reset or safe position shall be placed in the test chamber and the temperature shall be lowered to -40°F + 5°F and stabilized (see 6.4). This temperature shall be maintained for 4 hours minimum. After returning the unit to room temperature, test in accordance with 4.4.1, 4.4.2.1, and 4.4.4. No forced warming shall be used to return the unit to ambient room temperature.

4.4.10 High temperature. The unit in the reset or safe position shall be placed in the test chamber and the temperature raised to 125°F + 5°F and stabilized (see 6.4). This temperature shall be maintained for 4 hours minimum. After returning the unit to room temperature, test in accordance with 4.4.1, 4.4.2.1, and 4.4.4. No forced cooling shall be used to return the unit to ambient room temperature.

4.4.11 Temperature shock. The unit in the reset or safe position shall be placed in a test chamber at 125°F + 5°F and maintained at the temperature for a minimum of 4 hours. The unit shall then be removed from the high temperature chamber, and, within one minute, placed in a chamber with an in-

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ternal temperature of -40°F + 5°F and maintained at the temperature for a minimum of 4 hours. Remove the unit from the low temperature chamber and within 1 minute return the unit to the high temperature chamber. This sequence constitutes one cycle. Three total cycles shall be executed. Return the unit to room temperature and test in accordance with 4.4.1, 4.4.2.1, 4.4.4.1, and 4.4.3 and examine for damage (see 6.5).

4.4.12 Life tests. The unit shall be functioned as specified in 4.4.4.1. The unit shall be tested in accordance with 4.4.2.1 after every 10th set cycle, reset (to complete testing per 4.4.2.1), and then tested per 4.4.4.2. The set operation produced by 4.4.4.2 shall be counted as one of the next 10 cycles. The rate at which the unit is cycled shall not exceed one cycle (reset-set-reset) per minute.

4.4.13 Back emf. Assemble equipment as shown in Figure 3. The motor leads are directly connected to the input of the integrator. Turn the motor shaft through a known angle (300° minimum). The integrator will sum the motor back EMF for this angle during shaft rotation. The final output of the integrator after the motor shaft has been rotated through a fixed angle is a constant no matter what rate the motor shaft is turned. The integrator output has become independent of shaft velocity. The integrator output may be displayed on any high accuracy display. A sample calculation for a good quality motor follows:

Test fixture motor shaft angle = 300°

Digital display device reading = 115 millivolts

Integrator τ = 1 second: (RXC)

$$\theta = 300° \times \frac{2\pi}{360°} = 5.24 \text{ radians}$$

$$K_B = \frac{\text{DVM reading} \times \tau}{\theta} = \frac{(115\text{MV}) \times (1 \text{ sec})}{5.24 \text{ radians}}$$

K_B = 0.020 volts/radian/sec minimum

4.4.13.1 Effectively, the motor is tested as a tachometer generator and the time integrated angular output voltage is measured to determine quality of torque sensitivity.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The switching unit shall be packaged in accordance with dwg 9278227 using the specified moisture barrier.

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5.1.2 Level C. The switching unit shall be packaged in accordance with dwg 9278227 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The switching unit shall be packed in accordance with dwg 9278227.

5.3 Marking. Marking shall be as specified in accordance with dwg 9278227.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.2.1). Data specified in 3.5 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Physical damage. Physical damage shall include bent or loose contact pins, split or cracked insulation, damaged seal interfaces or cracks in body.

6.6 Visual inspection. A 4X minimum magnifying glass shall be used to perform this inspection.

6.7 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2307A dated 15 April 1975 and SCN No. 1 dated 5 June 1975.

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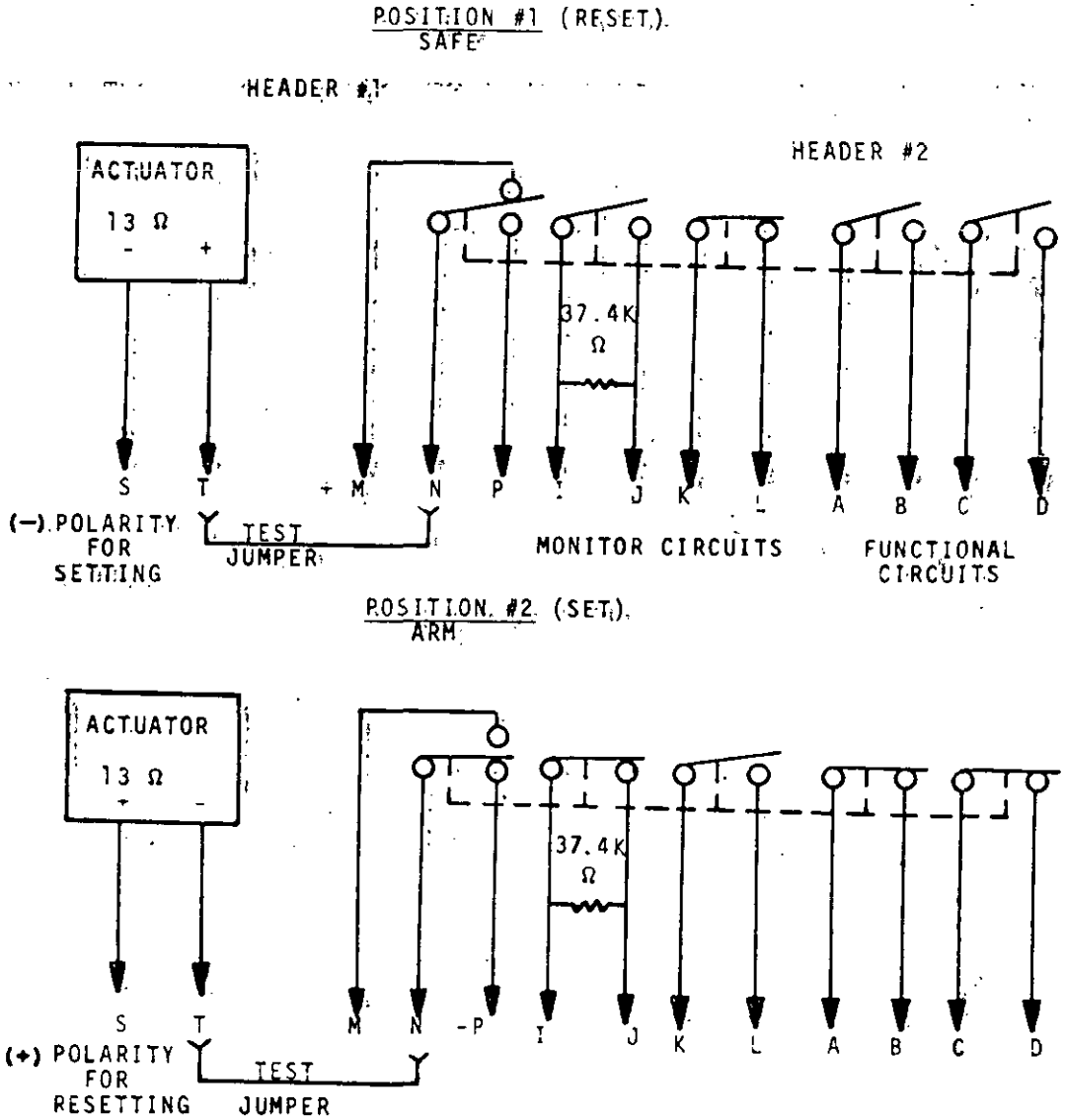
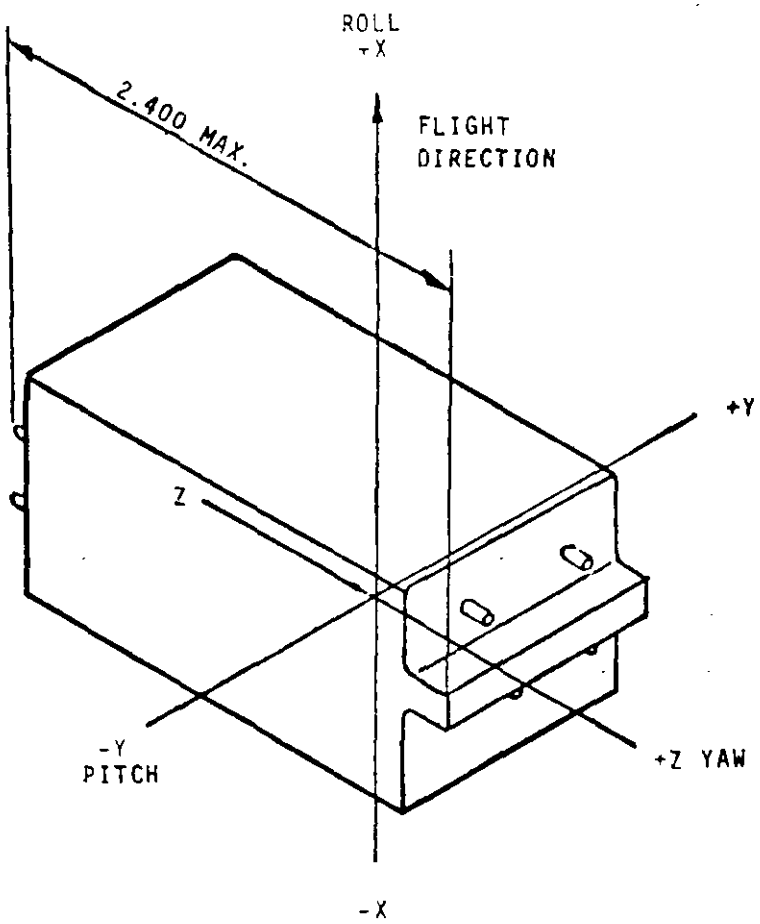


Figure 1.

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NOTE: ALL DIMENSIONS IN INCHES

Figure 2. XM70

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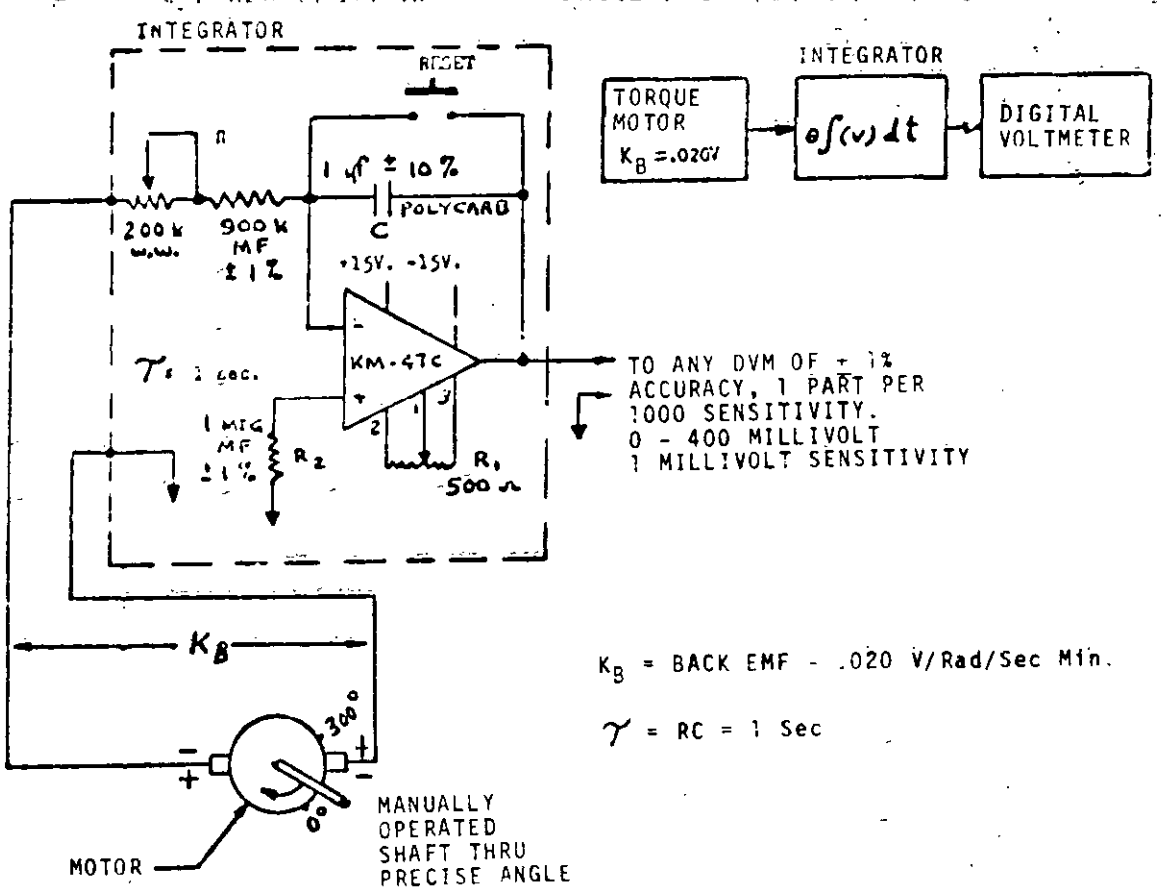


Figure 3. Motor Test Schematic

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